

COMPREHENSIVE CHEST CT ANALYSIS

Ensemble AI • Nodule & PE Assessment • Research Edition



Patient ID	40de7758-45e8-47ee-8adc-69b051d0461d	Study Date	2026-06-16
Name	Sam	Study Time	10:19:39
Age	24	Modality	CHEST CT
Sex	Male	Report ID	40de7758

AI ANALYSIS

Radiologist review required. This AI analysis is provided for clinical decision support and should be confirmed by a qualified radiologist.

Acquisition detected: Non Contrast • routed to 'nodule' pathway

AI TRIAGE CLASSIFICATION

AI triage flag: possible Lung Mass pattern on single-slice CT image • Model Confidence: 99.9% • Radiologist review recommended

Triage tier based on ACR Actionable Findings categories. AI suggestion — requires radiologist confirmation.

Single-slice DICOM analysis only — not a complete CT chest study interpretation.

DIFFERENTIAL DIAGNOSES

Rank	Finding	Confidence	ICD-10
1	Lung Mass	99.9%	R91.8
2	Pneumonia	93.1%	J18.9
3	Normal	3.5%	Z00.00
4	Nodule	2.3%	R91.1
5	PE	0.0%	I26.99

Findings are scored independently and are not mutually exclusive; multiple findings may show high confidence at the same time.

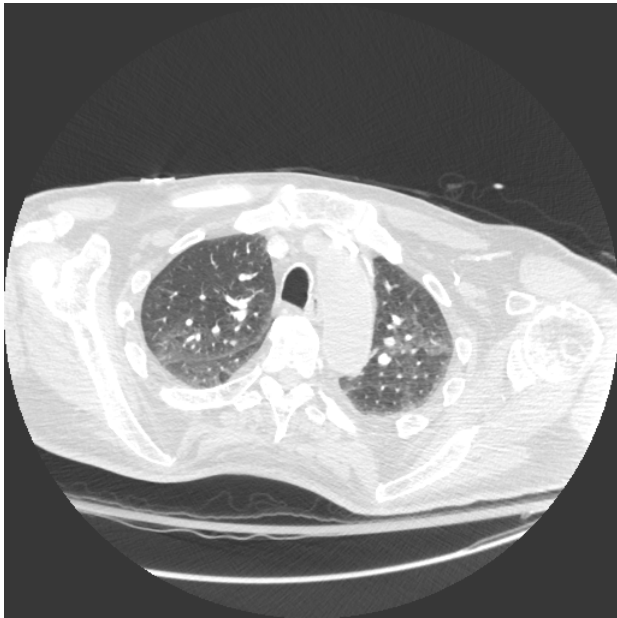
QUANTITATIVE MEASUREMENTS

Quantitative analysis unavailable for this scan

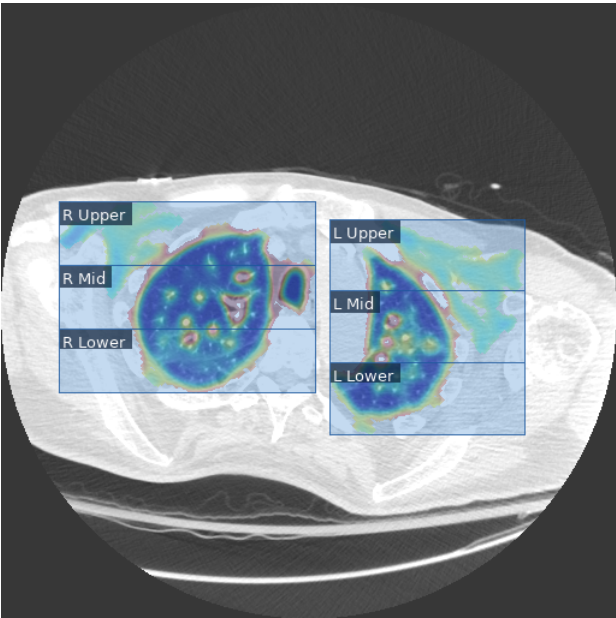
|| LUNG-RADS CLASSIFICATION

Lung-RADS Category	N/A
Malignancy Risk	N/A
Management	Radiologist review required

|| AI ANALYSIS VISUALIZATION



Original Scan



Analysis Overlay (Illustrative)

Illustrative overlay derived from image contrast — not AI-derived attention and not diagnostic. A validated AI attention map is shown once the model is validated. Radiologist review required.

Zone boxes mark lung regions (R/L, upper/middle/lower) for localization, not AI findings.

|| CLINICAL RECOMMENDATIONS

Priority Level	Primary Action	Follow-up
ROUTINE	Radiologist review required	Per radiologist

PER-MODEL CONFIDENCE BREAKDOWN

Single-pass inference — per-fold breakdown not applicable for this run.

Model	Probability
Pneumonia	93.1%
Normal	3.5%
Nodule	2.3%
LungMass	1.0%
PE	0.0%

CALIBRATION DIAGNOSTICS

Metric	Value
Shown confidence (to clinician)	93.1%
Raw ensemble agreement	93%
Safety floor (72%) applied	No
Indeterminate flag	No

CLINICAL SCORING

AI Pulmonary Risk Tier	N/A
Smoking history (clinical input)	Requires clinical input

IMPORTANT DISCLAIMER

- This AI report is for decision support only.
- All findings must be verified by a board-certified radiologist.

II TECHNICAL APPENDIX & NOMENCLATURE

This appendix contains reference tables used by the AI system. These tables are for clinical, academic, and validation reference and apply to all CT Chest studies.

Lung-RADS v2022 Category Reference

Category	Descriptor	Malignancy Risk	Management
1	No nodules / definitely benign	<1%	Continue annual screening (12 mo)
2	Benign appearance/behavior	<1%	Continue annual screening (12 mo)
3	Probably benign	1–2%	6-month LDCT
4A	Suspicious	5–15%	3-month LDCT or PET/CT
4B	Suspicious	>15%	Chest CT ± PET/CT, tissue sampling
4X	Suspicious + additional features	>15%	As 4B, expedited workup

Fleischner Society 2017 — Incidental Solid Nodule Follow-up

Nodule Size	Low Risk	High Risk
<6 mm	No routine follow-up	Optional CT at 12 mo
6–8 mm	CT at 6–12 mo	CT at 6–12 mo, then 18–24 mo
>8 mm	Consider CT at 3 mo, PET/CT	CT at 3 mo, PET/CT or biopsy

CTPA / Pulmonary Embolism Severity Reference

Clot Burden	Severity	Typical Action
Subsegmental	Low	Clinical correlation; consider anticoagulation
Segmental	Moderate	Anticoagulation; risk stratify
Central / saddle	High	Immediate evaluation; consider thrombolysis

ICD-10 Codes for Common Thoracic Findings

Finding	ICD-10	Description
Pulmonary embolism	I26.99	Other pulmonary embolism without acute cor pulmonale
Pneumonia	J18.9	Pneumonia, unspecified organism
Lung nodule	R91.1	Solitary pulmonary nodule
Lung mass	R91.8	Other nonspecific abnormal finding of lung field
Normal study	Z00.00	Encounter for general exam without abnormal findings

For internal validation / research use — not part of the standard clinical report.

Not autonomous diagnosis · not a replacement for clinical judgment · not for life-threatening emergency triage · not for direct-to-patient interpretation without clinician mediation · not for unqualified personnel · not for any modality outside the validated list.